Tianyi Zhang

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EDUCATION

Georgia Institute of Technology Atlanta, GA
Master of Science in Computer Science *Aug 2025 - May 2027*

Emory University Atlanta, GA
Bachelor of Science in Quantitative Sciences *Aug 2021 - May 2025*

PROFESSIONAL EXPERIENCE

LinkedIn Corporation Mountain View, CA
AI/ML Engineer Intern | Feed AI, LLM Retrieval *May 2026 - Present*

- Built a backend-agnostic **Adaptive User Representation Layer** that emits compact short-/long-term intents, lifecycle state, confidence, and evidence provenance as shared conditioning for ANN, GR, SID, or larger retrieval models.
- Built a production-scale **PySpark scorecard over 2.7B member-candidate pairs** and matched lifecycle evaluation; uncovered **+25.1% Recall@10 / +26.8% NDCG@10** but **21.6% lower ILD** for Semi-cold versus Active members, informing model selection and lifecycle strategy that balance relevance with feed breadth.
- Improved distributed training and serving reliability through layered configuration, reproducible execution, train/serve parity checks, serving observability, and one-axis-at-a-time validation across representation, model capacity, retrieval backend, and lifecycle policy.

Georgia Tech Efficient & Intelligent Computing Lab - NVIDIA Collaboration Atlanta, GA
Student Researcher | LLM Evaluation, Agentic AI *Jul 2025 - Dec 2025*

- Co-authored **AgentSuite (ICML 2026)**, auditing validity flaws in LLM-agent benchmarks; matched expert judgments at **0.79-0.87 F1** across 6 benchmarks and showed flaws affect **~23% of tasks**, reshuffling **63% of model rankings**.
- Built taxonomy-guided LLM-judge and rule-based structural detectors; generalized to unseen benchmarks without retuning and surfaced **100+** undocumented issues.

Emory Graph Mining Lab Atlanta, GA
Machine Learning Researcher | Graph Learning, AI for Science *Dec 2023 - May 2025*

- First-authored a causal brain-connectivity method (**AIME 2025**) using Granger-directed graphs to improve clinical prediction across **~8K patients**.
- Co-authored a state-based Transformer (**IEEE EMBC 2025, oral**) modeling temporal transitions across latent functional-connectivity states.

PUBLICATIONS

H. Suh*, S. Lee*, B. Ji*, et al., including **T. Zhang**. "AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline." *ICML* 2026.

T. Zhang, K. Han, J. Nie, et al. "Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis." *AIME* 2025.

J. Nie, K. Han, **T. Zhang**, et al. "Brain Network State Transformer." *IEEE EMBC* 2025. (Oral)

PROJECTS & OPEN SOURCE

CoAct: Governance Substrate for Multi-Agent Coding - Dependency-free Go binary for capability policies, locks, worktree isolation, merge gates, and append-only audit journals.

NVIDIA NeMo RL: Open-Source Contributor - Contributions to a scalable reinforcement-learning and LLM post-training library.

TECHNICAL SKILLS

Programming: Python, Go, SQL, Java, R, C/C++ **ML/AI:** PyTorch, CUDA, Hugging Face, vLLM, SGLang, MLflow
Infra/Data: Spark/PySpark, Docker, AWS, GitHub Actions, Linux, PostgreSQL, MongoDB, Redis